

5.9.1

RAVULA SHASHANK REDDY - EE25BTECH11047

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Question

A fraction becomes $\frac{1}{3}$ when 1 is subtracted from its numerator and it becomes $\frac{1}{4}$ when 8 is added to its denominator. Find the fraction.

Solution

Let fraction be $\frac{x}{y}$

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

$$\frac{x-1}{y} = \frac{1}{3} \quad (2)$$

$$\begin{pmatrix} 3 \\ -1 \end{pmatrix}^T \mathbf{x} = 3 \quad (3)$$

$$\frac{x}{y+8} = \frac{1}{4} \quad (4)$$

$$\begin{pmatrix} 4 \\ -1 \end{pmatrix}^T \mathbf{x} = 8 \quad (5)$$

$$\begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 3 \\ 8 \end{pmatrix} \quad (6)$$

Solution

$$\begin{pmatrix} 3 & -1 & 3 \\ 4 & -1 & 8 \end{pmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{3}R_1} \begin{pmatrix} 1 & -\frac{1}{3} & 1 \\ 4 & -1 & 8 \end{pmatrix} \quad (7)$$

$$R_2 \rightarrow R_2 - 4R_1 \Rightarrow \begin{pmatrix} 1 & -\frac{1}{3} & 1 \\ 0 & \frac{1}{3} & 4 \end{pmatrix} \quad (8)$$

$$R_2 \rightarrow 3R_2 \Rightarrow \begin{pmatrix} 1 & -\frac{1}{3} & 1 \\ 0 & 1 & 12 \end{pmatrix} \quad (9)$$

$$R_1 \rightarrow R_1 + \frac{1}{3}R_2 \Rightarrow \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & 12 \end{pmatrix} \quad (10)$$

$$\mathbf{x} = \begin{pmatrix} 5 \\ 12 \end{pmatrix} \quad (11)$$

$$\text{Required Fraction} = \frac{5}{12}$$

```
#include <stdio.h>

int main() {
    // Augmented matrix [A|b]
    double mat[2][3] = {
        {3, -1, 3}, //  $3x - y = 3$ 
        {4, -1, 8} //  $4x - y = 8$ 
    };

    // Row operation:  $R2 \rightarrow R2 - (4/3)R1$ 
    double factor = mat[1][0] / mat[0][0];
    for (int j = 0; j < 3; j++) {
        mat[1][j] = mat[1][j] - factor * mat[0][j];
    }
}
```

```
// Scale R2 to make pivot = 1
double scale = mat[1][1];
for (int j = 0; j < 3; j++) {
    mat[1][j] /= scale;
}

// Back substitution
double y = mat[1][2];
double x = (mat[0][2] - mat[0][1] * y) / mat[0][0];

printf(x = %.2f\n, x);
printf(y = %.2f\n, y);
printf(Fraction = %.2f/%.2f\n, x, y);

return 0;
}
```

Python Direct

```
import numpy as np
import libs.line.funcs as line
import libs.triangle.funcs as tri

# Coefficient matrix A and RHS vector b
A = np.array([[3, -1],
              [4, -1]], dtype=float)
b = np.array([3, 8], dtype=float)

# Solve system A x = b
x = np.linalg.solve(A, b)

num, den = x[0], x[1]

print(x =, num)
print(y =, den)
print(Fraction = {}/{}.format(int(num), int(den)))
```

```
import numpy as np
import ctypes

# Load shared object
solver = ctypes.CDLL('./solver.so')

# Define return types and arguments
solver.solve.argtypes = [np.ctypeslib.ndpointer(dtype=np.float64,
        ndim=1, flags=C)]

# Prepare output array
out = np.zeros(2, dtype=np.float64)
```



```
# Call C function
solver.solve(out)

x, y = out[0], out[1]

print(x =, x)
print(y =, y)
print(Fraction = {}/{}.format(int(x), int(y)))
```